



AJKD Atlas of Renal Pathology: Tip Lesion Variant of Focal Segmental Glomerulosclerosis

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Clinical and Pathologic Features

The tip lesion variant of focal segmental glomerulosclerosis (FSGS) is associated with nephrotic syndrome and usually responds to steroid therapy. This FSGS variant has the lowest risk for progression to kidney failure.

Light microscopy: Segmental glomerulosclerosis or endocapillary hypercellularity (usually foam cells) involving <50% of the glomerular tuft at the tubular pole, with an adhesion or confluence with tubular epithelium to the tubular outlet. Collapsing and perihilar variants must be excluded.

Immunofluorescence microscopy: No or limited deposits (nonspecific immunoglobulin M and C3 staining in sclerotic areas).

Electron microscopy: Extensive foot process effacement. No or limited deposits.

Etiology/Pathogenesis

The etiology and pathogenesis of tip lesion has not yet been defined. Injury to podocytes by turbulent flow at the tubular pole has been proposed.

Differential Diagnosis

Usually of primary etiology, but may be seen in other diseases with heavy proteinuria (eg, membranous nephropathy).

Key Diagnostic Features

- Extensive foot process effacement
- Absence of immune complexes
- Diagnostic tip lesion without collapsing or perihilar variants

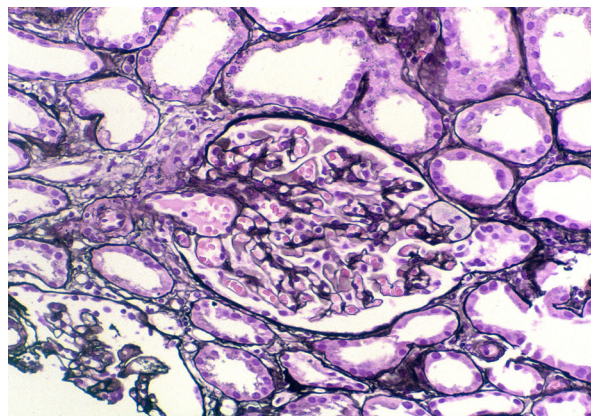


Figure 1. Tip lesion with localized sclerotic lesion only involving the proximal tubular pole of the glomerulus (Jones silver stain). Reproduced with permission from *AJKD* 38(6):e30.

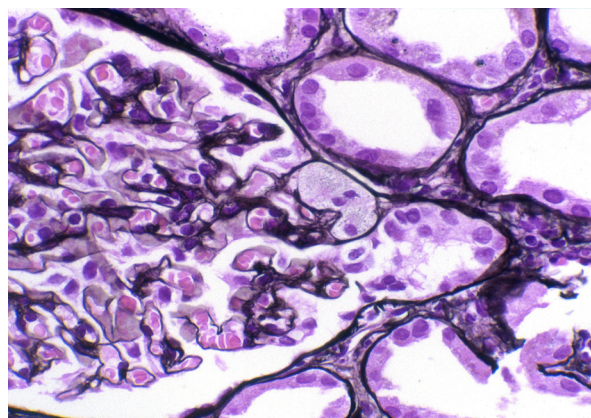


Figure 2. Sclerotic lesion localized to the proximal tubular pole with adhesion of the capillary loop to Bowman capsule and intra-capillary foam cells, characteristic of the tip lesion variant of focal segmental glomerulosclerosis (Jones silver stain). Reproduced with permission from *AJKD* 38(6):e30.

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